03603332 OPERATING SYSTEMS · GROUP PROJECT · DEADLOCKS

# Deadlock Detector

Find deadlocks without changing the code.

```
$ LD_PRELOAD=./build/libdetect.so ./your_program
```

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THE PROBLEM — A BANK TRANSFER

# Two threads. Two locks. Stuck forever.

**Thread 1**
moves money Alice → Bob

| | | |
|---|---|---|
| t0 | lock(Alice) | ✓ got it |
| t2 | lock(Bob) | ⏳ waits |

**Thread 2**
moves money Bob → Alice

| | | |
|---|---|---|
| t1 | lock(Bob) | ✓ got it |
| t3 | lock(Alice) | ⏳ waits |

## No crash. No error. 0% CPU. It just stops.

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DEADLOCK BASICS

# 4 Coffman conditions

01

## Mutual exclusion

Only one thread can hold a lock.

02

## Hold and wait

Each thread holds one account. It wants the other.

03

## No preemption

Nobody can take a lock away.

04

## Circular wait

Thread 1 waits for Thread 2. Thread 2 waits for Thread 1.

**All 4 true → deadlock. Break one → no deadlock.**

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OUR GOAL

# Make the silent program speak.

✓ Which threads

✓ Which locks

✓ Which code lines

No code change

No rebuild

```
$ LD_PRELOAD=./build/libdetect.so ./your_program
```

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BIG PICTURE

# Four steps

STEP 1

## Catch

every lock call

`interpose.c`

→

STEP 2

## Record

who holds, who waits

`state.c`

→

STEP 3

## Check

for a loop every 500 ms

`detector.c`
`graph.c`

→

STEP 4

## Report

threads, locks, code lines

`report.c`

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STEP 1 — CATCH

# LD_PRELOAD puts us in front

**Your program**

```
pthread_mutex_lock(&alice.lock)
```

↓

**libdetect.so — our library**

```
1  note "waiting for Alice"
2  call the real lock
3  note "got Alice"
```

↓

**libc.so.6 — the real lock**

↓

**Linux kernel — we do not touch it**

## LD_PRELOAD loads our library first.

So our lock function runs before the real one.

- `pthread_mutex_lock`
- `pthread_mutex_trylock`
- `pthread_mutex_unlock`

STEP 2 — RECORD

# Two tables

## holds[]

Who holds each lock?

| LOCK | HELD BY |
|---|---|
| Alice | Thread 1 |
| Bob | Thread 2 |

## waits[]

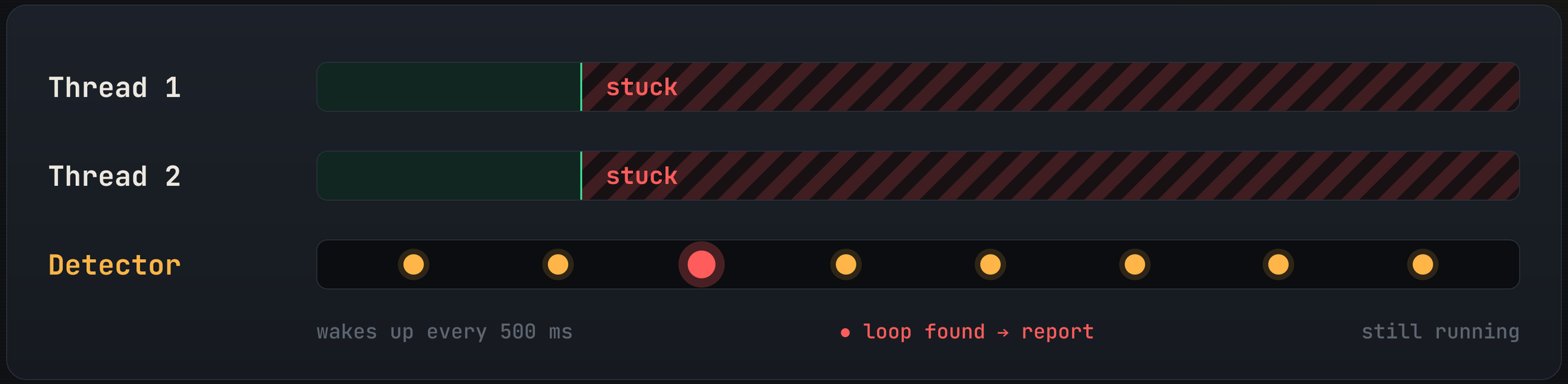
Who waits for which lock?

| THREAD | WAITS FOR |
|---|---|
| Thread 1 | Bob |
| Thread 2 | Alice |

Fixed size. **No malloc.** ← malloc uses locks inside → it would call our code again → `in_dd` flag stops the loop

STEP 3 — CHECK

# The detector thread




OWN THREAD

**It copies the tables and looks for a loop.**

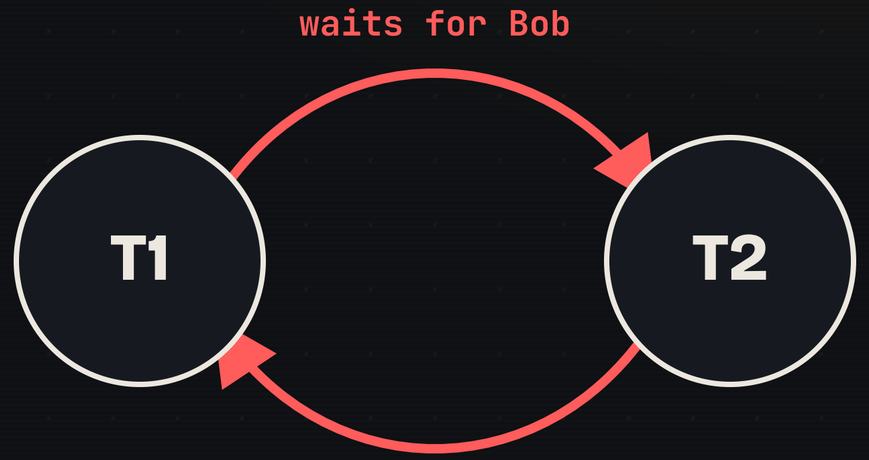
NEVER STUCK

**It never uses the program's locks.**

STEP 3 — CHECK

# Wait-for graph: find the loop

| TIME | WHAT HAPPENS | GRAPH |
|---|---|---|
| t0 | T1 locks Alice | Alice held by T1 |
| t1 | T2 locks Bob | Bob held by T2 |
| t2 | T1 wants Bob → waits | T1 → T2 |
| t3 | T2 wants Alice → waits | T2 → T1 |
| ≤ 500 ms | detector finds the loop → report | |




loop = circular wait = deadlock

One lock at a time → **max one arrow out** → just follow the arrows. finds every loop · up to 16

STEP 4 — REPORT

# The real output

```
[DEADLOCK DETECTED] 2 threads in cycle

  Thread 514
      holds   mutex 0x56fc6b20b070   at bank.c:89
      waiting mutex 0x56fc6b20b030   at bank.c:94
  Thread 513
      holds   mutex 0x56fc6b20b030   at bank.c:65
      waiting mutex 0x56fc6b20b070   at bank.c:70

  Cycle: T514 → T513 → T514
  Hint : acquire locks in a consistent global order
```

**holds** got this lock at this line

**waiting** stuck at this line

**Cycle** the loop

**Hint** how to fix it

**514 / 513** thread IDs from Linux

code lines: dladdr + addr2line · needs -g and PIE

DD_DOT_OUT → picture (Graphviz)

HOW WE TESTED

# Tests that must pass

## Must deadlock + report

**bank**
2 threads, money transfer

**sample_webapp**
looks like real code

**cycle3**
3 threads in one loop

**two_pairs**
2 loops at once

## Must stay quiet

**no_deadlock**
same lock order → no report

**condvar**
condition variable → no report

## Hard cases

**robust**
lock owner died

**errcheck**
wrong thread unlocks

**longcycle**
70 threads in one loop

**lockorder**
opposite order, no deadlock → warning

PTHREAD_BARRIER, NOT SLEEP

**The deadlock happens every time.**

MAKE REGRESS

**Each test catches its bug if we remove the fix.**

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# Live demo

**1**

bank alone stops forever

`bank → exit 124`

**2**

same command + LD_PRELOAD ⟶ report

`LD_PRELOAD=./build/libdetect.so`

**3**

picture of the loop

`DD_DOT_OUT=cycle.dot`

**4**

same lock order ⟶ finishes, no report

`no_deadlock → exit 0`

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RESULTS

# It works. It has a cost.

| PROGRAM | WITHOUT TOOL | WITH LD_PRELOAD |
|---|---|---|
| bank | hangs | hangs + report |
| sample_webapp | hangs | hangs + report |
| cycle3 | hangs | report: 3 threads |
| two_pairs | hangs | 2 reports, 1 per loop |
| no_deadlock | exit 0 | exit 0, no report |
| lockorder | exit 0 | exit 0 + order warning |

**20 / 20**

runs found the deadlock
(bank and sample_webapp)

cost per lock + unlock

| | |
|---|---|
| 1 thread | 2.3 → 72.1 ns ×31.3 |
| 4 threads, 1 lock | 39.9 → 313.5 ns ×7.9 |

**Good for finding bugs.** Too slow to leave on all the time.

Speaker D

LIMITS & EARLY WARNING

# What it can and cannot do

LIMIT

## Linux, normal mutex

Linux programs only. No condition variables, rwlock or semaphore. No livelock.

LIMIT

## Finds it after

A deadlock is found after it happens. It does not stop it.

ALSO

## Warns early

Two locks taken in opposite orders → warning before a deadlock.

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SUMMARY

# Three things to remember

1. A deadlock is a **loop** in the wait-for graph.
2. Our tool finds it with **no code change** and shows the exact lines.
3. The fix: **lock in the same order.**

**DETECTION**
this tool

**PREVENTION**
same lock order (demo step 4) + early warning

**AVOIDANCE**
our Banker's simulator · docs/banker.html

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try it: docs/playground.html

DEADLOCK DETECTOR

# Thank you.
# Questions?

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★ Everyone